Alcohols, Phenols and Ethers

Alcohols, phenols and ethers are the parent compounds behind detergents, antiseptics and fragrances, in that order.

By the end of this unit

- name alcohols, phenols and ethers by the IUPAC system
- set out the reactions by which an alkene, an aldehyde, a ketone or a carboxylic acid becomes an alcohol
- set out the reactions by which a haloarene, a benzene sulphonic acid, a diazonium salt or cumene becomes a phenol
- set out the reactions that make ethers from (i) alcohols and (ii) alkyl halides with sodium alkoxides or aryloxides
- tie the physical properties of the three families to their structures
- read the chemical reactions of all three off their functional groups

Where the three families turn up

- replacing one or more hydrogen atoms of a hydrocarbon by another atom or group builds an **entirely new compound**, with properties and uses of its own
- put **–OH** in place of a hydrogen in an aliphatic hydrocarbon and an **alcohol** results; do it to an aromatic one and the result is a **phenol**
- the **spirit** that polishes wooden furniture is chiefly one hydroxyl compound, **ethanol**
- **sugar** on the table, **cotton** in fabric, **paper** under the pen: all are built of compounds carrying –OH groups
- without paper there would be no note-books, books, newspapers, currency notes, cheques or certificates, and no magazines carrying photographs and stories

What each name covers

- an **alcohol** holds one or more **hydroxyl (OH) groups bonded straight to carbon** in an **aliphatic** system, as CH_3OH does
- a **phenol** holds **–OH bonded straight to carbon** of an **aromatic** system, as C_6H_5OH does
- put an **alkoxy or aryloxy group (R–O / Ar–O)** in place of a hydrogen of a hydrocarbon and the product belongs to the **ethers** — CH_3OCH_3, dimethyl ether, for one

The unit runs in two parts: **I · Alcohols and phenols**, then **II · Ethers**.

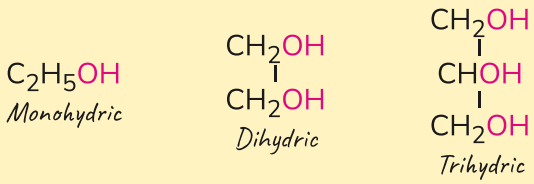
- an ether may also be pictured as what results when the **hydrogen of the hydroxyl group** of an alcohol or a phenol is **replaced by an alkyl or aryl group**
- this unit takes the chemistry of the three classes together: **alcohols, phenols and ethers**

A.1 Classification

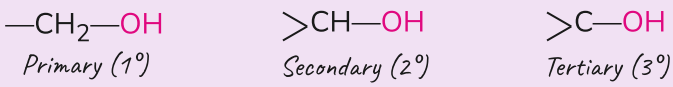
- classifying compounds makes their study **systematic and simpler**, so begin with how alcohols, phenols and ethers are sorted

Alcohols — mono, di, tri or polyhydric

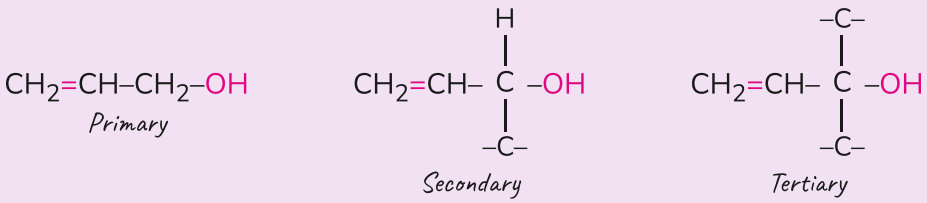
- alcohols and phenols count as **mono–, di–, tri– or polyhydric** according to whether they hold **one, two, three or many hydroxyl groups**



- a monohydric alcohol is sorted further by the **hybridisation of the carbon the hydroxyl group sits on**
- **(i) Compounds with a C_{sp3}–OH bond** — here the –OH is on an ***sp*^{3} hybridised carbon of an alkyl group**, and these divide again
- **Primary, secondary and tertiary alcohols** — the –OH sits on a primary, a secondary or a tertiary carbon atom

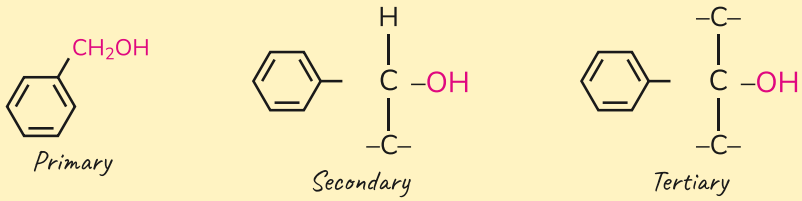


- **Allylic alcohols** — the —OH sits on an ***sp*^{3} carbon next to a carbon-carbon double bond**, that is, on an **allylic carbon**



- **Benzylic alcohols** — the —OH sits on an ***sp*^{3} carbon standing next to an aromatic ring**; examples follow on sheet 3

Benzylic alcohols · the three kinds



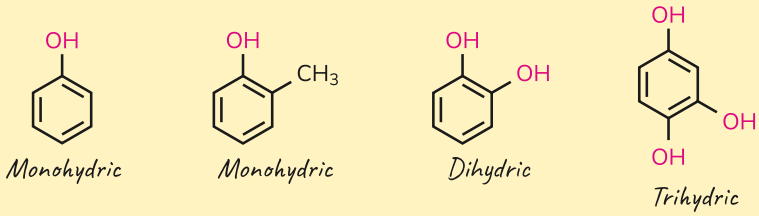
- an allylic or a benzylic alcohol may itself be **primary, secondary or tertiary**

- **(ii) Compounds with a C_{sp2}–OH bond** — the —OH is bonded to a carbon of a **carbon-carbon double bond**, that is to a **vinylic carbon**, or to an **aryl carbon**; these are also called **vinylic alcohols**

$$CH_2 = CH - OH$$

Vinylic alcohol

Phenols — mono, di and trihydric phenols



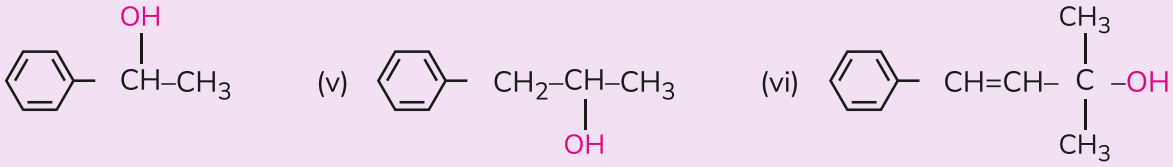
Ethers

- an ether counts as **simple** or **symmetrical** when the two alkyl or aryl groups on the oxygen are the **same**, and as **mixed** or **unsymmetrical** when they **differ**
- diethyl ether, $C_2H_5OC_2H_5$, is symmetrical; $C_2H_5OCH_3$ and $C_2H_5OC_6H_5$ are unsymmetrical

Intext questions

Q1 Sort each of these into primary, secondary and tertiary alcohols.

(i) $CH_3-C(CH_3)_2-CH_2OH$ (ii) $H_2C = CH- CH_2OH$ (iii) $CH_3- CH_2- CH_2- OH$



Q2 Which of the examples above are allylic alcohols?

A.2 Nomenclature

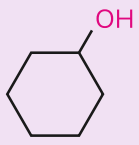
- **(a) Alcohols** — for the **common name**, take the alkyl group's own common name and put the word **alcohol** after it: CH_3OH is methyl alcohol

- the **IUPAC name** of an alcohol comes from the **alkane it derives from**, with the **'e' of the alkane replaced by the suffix 'ol'**
- substituent positions are given by **numerals**: number the **longest carbon chain**, the parent chain, from the end **nearest the hydroxyl group**
- the –OH and every other substituent then takes the number of the carbon it is attached to
- for a **polyhydric** alcohol the alkane keeps its 'e' and 'ol' is added, with a **multiplicative prefix** — di, tri and so on — before 'ol'
- locants mark where the **–OH** groups sit: HO–CH_2–CH_2–OH is **ethane–1, 2-diol**
- Table 7.A sets out common and IUPAC names for a few alcohols

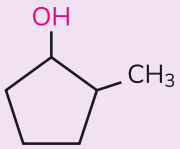
Table 7.A **Common and IUPAC Names of Some Alcohols**

Compound	Common name · IUPAC name
CH_3-OH	Methyl alcohol Methanol
$CH_3-CH_2-CH_2-OH$	*n*-Propyl alcohol Propan-1-ol
$CH_3-\underset{\displaystyle OH}{\underset{\vert}{CH}}-CH_3$	Isopropyl alcohol Propan-2-ol
$CH_3-CH_2-CH_2-CH_2-OH$	*n*-Butyl alcohol Butan-1-ol
$CH_3-\underset{\displaystyle OH}{\underset{\vert}{CH}}-CH_2-CH_3$	*sec*-Butyl alcohol Butan-2-ol
$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-CH_2OH$	Isobutyl alcohol 2-Methylpropan-1-ol
$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{\overset{\displaystyle CH_3}{\overset{\vert}{C}}}}-OH$	*tert*-Butyl alcohol 2-Methylpropan-2-ol
$HO-H_2C-CH_2-OH$	Ethylene glycol Ethane-1,2-diol
$CH_2(OH)-CH(OH)-CH_2(OH)$	Glycerol Propane -1, 2, 3-triol

- a **cyclic alcohol** takes the prefix **cyclo**, and its —OH is counted as sitting on **C–1**

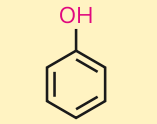


Cyclohexanol

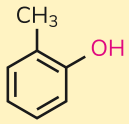


2-Methylcyclopentanol

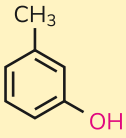
- **(b) Phenols** — benzene's simplest hydroxy derivative is **phenol**, a name that serves both as the common one and as an accepted IUPAC one
- a phenol is built round a benzene ring, so its substituted compounds carry the common-name prefixes ***ortho*** for 1,2-, ***meta*** for 1,3- and ***para*** for 1,4-disubstitution



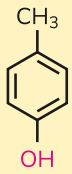
Phenol · Phenol



o-Cresol · 2-Methylphenol



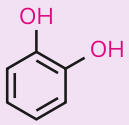
m-Cresol · 3-Methylphenol



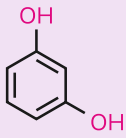
p-Cresol · 4-Methylphenol

common name above, IUPAC name below

- the **dihydroxy derivatives of benzene** go by the names **1, 2-, 1, 3- and 1, 4-benzenediol**



Catechol · Benzene-1,2-diol



Resorcinol · Benzene-1,3-diol

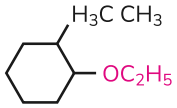


Hydroquinone or quinol · Benzene-1,4-diol

- **(c) Ethers** — a **common name** lists the alkyl or aryl groups as **separate words in alphabetical order** and closes with the word **ether**; $CH_3OC_2H_5$ is ethylmethyl ether

Table 7.B **Common and IUPAC Names of Some Ethers**

Compound	Common name · IUPAC name
CH_3OCH_3	Dimethyl ether Methoxymethane
$C_2H_5OC_2H_5$	Diethyl ether Ethoxyethane
$CH_3OCH_2CH_2CH_3$	Methyl *n*-propyl ether 1-Methoxypropane
$C_6H_5OCH_3$	Methyl phenyl ether (Anisole) Methoxybenzene (Anisole)
$C_6H_5OCH_2CH_3$	Ethyl phenyl ether (Phenetole) Ethoxybenzene

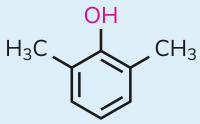
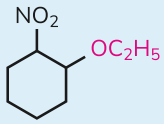
Compound	Common name · IUPAC name
$C_6H_5O(CH_2)_6-CH_3$	Heptyl phenyl ether 1-Phenoxyheptane
$CH_3O-CH(CH_3)-CH_3$	Methyl isopropyl ether 2-Methoxypropane
$C_6H_5-O-CH_2-CH_2-CH(CH_3)-CH_3$	Phenyl isopentyl ether 3-Methylbutoxybenzene
$CH_3-O-CH_2-CH_2-OCH_3$	— 1,2-Dimethoxyethane
	— 2-Ethoxy--1,1-dimethylcyclohexane

- when both alkyl groups match, the prefix **di** goes in front of the alkyl name: $C_2H_5OC_2H_5$ is diethyl ether
- under IUPAC an ether counts as a **hydrocarbon derivative** in which a hydrogen has been replaced by an **–OR** or **–OAr** group, R standing for alkyl and Ar for aryl
- the **larger (R) group is taken as the parent hydrocarbon**; Table 7.B gives a few worked names

Worked example 7.a

Give the IUPAC name of each compound.

(i) $CH_3–CH(Cl)–CH(CH_3)–CH(CH_3)–CH_2OH$ (ii) $CH_3–CH(CH_3)–O–CH_2CH_3$

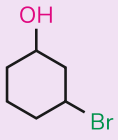
(iii)  (iv) 

- (i) 4-Chloro-2,3-dimethylpentan-1-ol · (ii) 2-Ethoxypropane
- (iii) 2,6-Dimethylphenol · (iv) 1-Ethoxy-2-nitrocyclohexane

Intext question

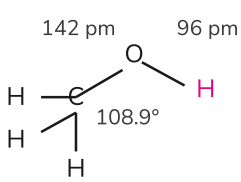
Q3 Give each compound below its IUPAC name.

(i) $CH_3–CH_2–CH(CH_2OH)–CH(CH_2Cl)–CH(CH_3)–CH_3$ (ii) $CH_3–CH(CH_3)–CH_2–CH(OH)–CH(CH_2OH)–CH_3$

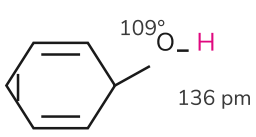
(iii)  (iv) $H_2C=CH–CH(OH)–CH_2–CH_2–CH_3$ (v) $CH_3–C(CH_3)=C(Br)–CH_2OH$

A.3 Structures of the functional groups

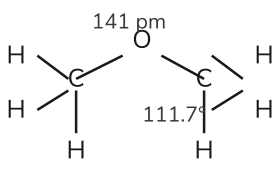
- in an alcohol the oxygen of the **–OH** group is joined to carbon by a **sigma (σ) bond**, made by the overlap of an ***sp*^{3} hybridised orbital of carbon** with an ***sp*^{3} hybridised orbital of oxygen**
- Figure 7.A sets out the structural detail for methanol, phenol and methoxymethane



Methanol



Phenol



Methoxymethane

Figure 7.A Structures of methanol, phenol and methoxymethane

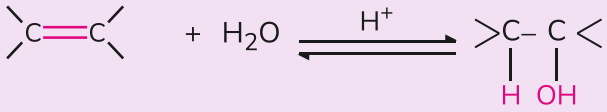
- the **C–O–H bond angle in an alcohol falls a little short of the tetrahedral angle** (109°-28'), because the **unshared electron pairs on oxygen repel one another**
- a phenol carries its –OH on an aromatic ring's ***sp*^{2} hybridised carbon**
- at **136 pm the C–O bond in phenol comes out a little shorter** than methanol's, for two reasons: it has **partial double bond character**, oxygen's unshared pair being conjugated with the ring (section A.6), and the carbon holding that oxygen is ***sp*^{2} hybridised**

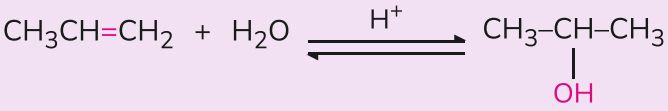
- in an ether the **four electron pairs on oxygen** — two bond pairs and two lone pairs — take up an approximately **tetrahedral arrangement**
- the bond angle runs a **little above the tetrahedral angle**, the two bulky (–R) groups repelling one another
- the **C–O bond length (141 pm)** is almost what it is in alcohols

A.4 Alcohols and phenols · preparation of alcohols

1 · From alkenes

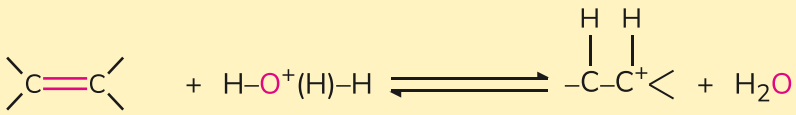
- **(i) By acid catalysed hydration** — an alkene takes up **water over an acid catalyst** and an alcohol results
- with an **unsymmetrical alkene** the addition follows **Markovnikov's rule**



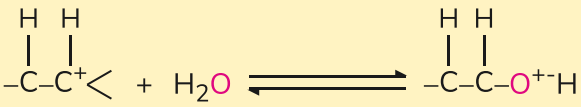


Mechanism · three steps

- **Step 1** — H_3O^+ attacks the alkene as an electrophile and **protonates it to a carbocation**: $H_2O + H^+ \rightarrow H_3O^+$



- **Step 2** — water attacks the carbocation as a **nucleophile**

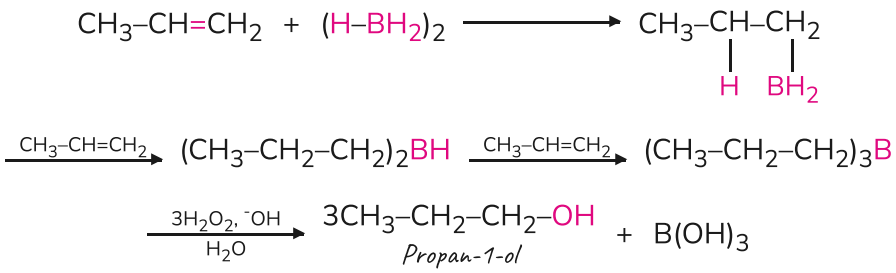


- **Step 3** — the intermediate **loses a proton**, and the alcohol is left



- **(ii) By hydroboration–oxidation** — **diborane $(BH_3)_2$** reacts with alkenes and the addition product is a **trialkyl borane**
- **hydrogen peroxide in aqueous sodium hydroxide** then oxidises that borane to the alcohol

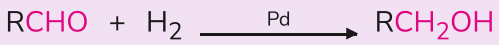
- **H.C. Brown** reported hydroboration-oxidation in **1959**; his work on organic compounds of boron won him a share of the **1979 Nobel prize in Chemistry, with G. Wittig**

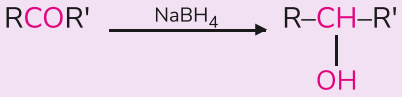


- borane adds across the double bond so that the **boron atom lands on the sp^2 carbon carrying the greater number of hydrogen atoms**
- the alcohol that follows looks as though **water had been added to the alkene against Markovnikov's rule**
- the yield of alcohol is **excellent**

2 · From carbonyl compounds

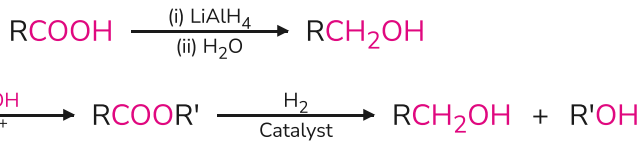
- **(i) Reducing an aldehyde or a ketone** — **hydrogen added over a catalyst** brings either down to the matching alcohol, in what is called **catalytic hydrogenation**
- the catalyst is usually a **metal in finely divided form**: platinum, palladium or nickel
- **sodium borohydride $(NaBH_4)$** and **lithium aluminium hydride $(LiAlH_4)$** do the same work
- an **aldehyde gives a primary alcohol**, a **ketone a secondary** one





- **(ii) By reduction of carboxylic acids and esters** — **lithium aluminium hydride**, a strong reducing agent, takes carboxylic acids down to **primary alcohols in excellent yield**
- $LiAlH_4$ is an **expensive reagent**, so it is kept for special chemicals
- industry reduces acids to alcohols by first **turning them into esters** (section A.6) and then **hydrogenating those over a catalyst**

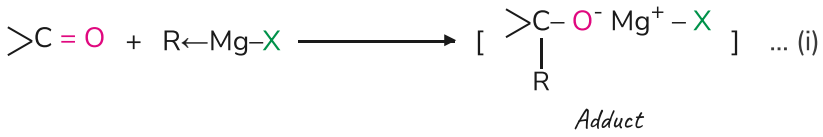
- numbers set in front of the reagents along an arrow mean the **second reagent goes in only once the first reaction is complete**

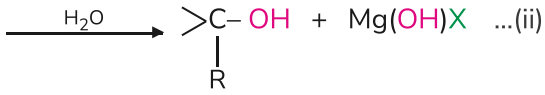


3 · From Grignard reagents

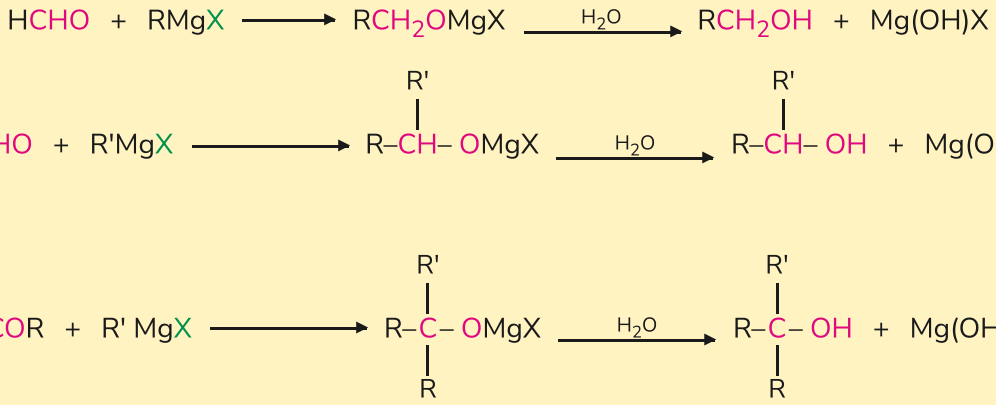
- alcohols come from **Grignard reagents** (Unit 6, Class XII) set against **aldehydes and ketones**
- the first step is **nucleophilic addition of the Grignard reagent to the carbonyl group**, giving an **adduct**
- **hydrolysing that adduct** then yields the alcohol

- a Grignard reagent with **methanal** gives a **primary** alcohol, with **other aldehydes** a **secondary** one, and with **ketones** a **tertiary** one





The overall reactions, aldehyde by aldehyde



Worked example 7.b

Give the structure and IUPAC name of the product expected from each reaction.

- (a) catalytic reduction of butanal
- (b) hydration of propene in dilute sulphuric acid
- (c) propanone with methylmagnesium bromide, followed by hydrolysis

(a) $CH_3–CH_2–CH_2–CH_2–OH$ — *Butan-1-ol*

(b) $CH_3–CH(OH)–CH_3$ — *Propan-2-ol*

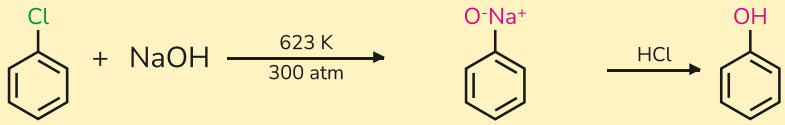
(c) $CH_3–C(CH_3)_2–OH$ — *2-Methylpropan-2-ol*

Preparation of phenols

- phenol, also called **carbolic acid**, was **first isolated from coal tar early in the nineteenth century**
- today it is **made commercially by synthesis**; in the laboratory it comes from benzene derivatives by the methods on sheet 10

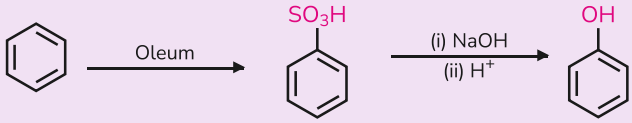
1 · From haloarenes

- **chlorobenzene fused with NaOH at 623 K and 320 atmospheric pressure** gives sodium phenoxide
- **acidifying that phenoxide** then yields phenol (Unit 6, Class XII)



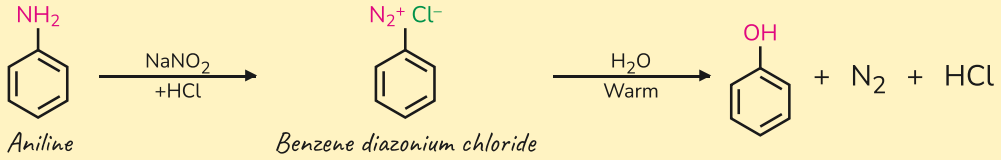
2 · From benzenesulphonic acid

- **oleum sulphonates benzene**; heat the **benzene sulphonic acid** that results with **molten sodium hydroxide** and sodium phenoxide follows
- **acidifying the sodium salt** gives phenol



3 · From diazonium salts

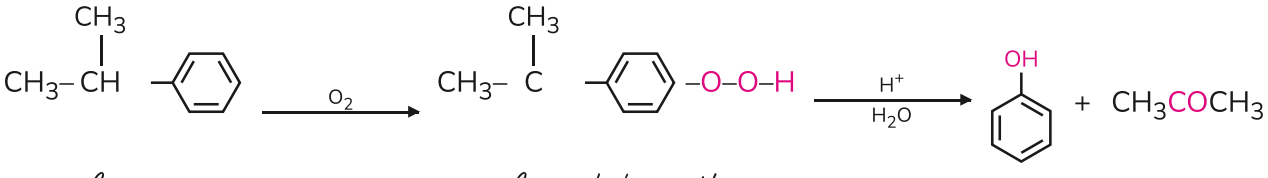
- a **diazonium salt** follows when an **aromatic primary amine** meets **nitrous acid ($NaNO_2$ + HCl) at 273-278 K**
- **warming with water hydrolyses a diazonium salt to a phenol**, and **dilute acid** does the same (Unit 9, Class XII)



4 · From cumene

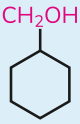
- the hydrocarbon **cumene is what phenol is manufactured from**
- cumene, that is **isopropylbenzene**, is **oxidised by air to cumene hydroperoxide**
- **dilute acid** then converts that to **phenol and acetone**
- **acetone**, the by-product, is itself obtained in large quantities this way

- most of the world's phenol is produced from cumene



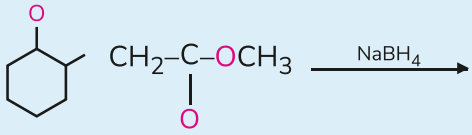
Intext questions

Q4 Show how each alcohol below may be made by setting a suitable Grignard reagent against methanal.

(i) $CH_3-CH(CH_3)-CH_2OH$ (ii) 

Q5 Draw the structures of the products of these reactions.

(i) $CH_3-CH=CH_2 \xrightarrow{H_2O/H^+}$

(ii) 

(iii) $CH_3-CH_2-CH(CH_3)-CHO \xrightarrow{NaBH_4}$

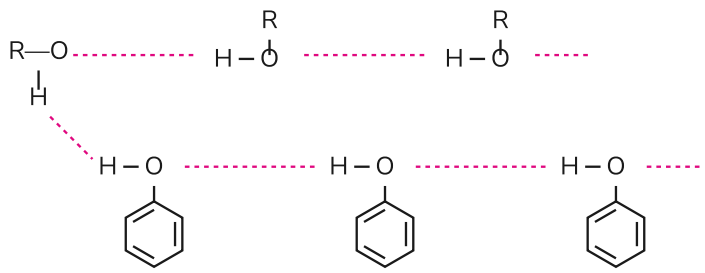
A.5 Physical properties

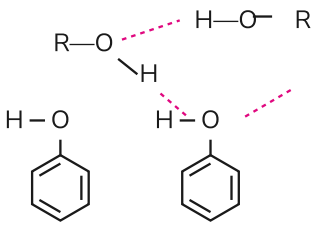
- an alcohol or a phenol is built of **two parts**: an **alkyl or aryl group** and a **hydroxyl group**
- the **hydroxyl group settles most of the properties**; the alkyl or aryl group only **modifies** them

Boiling points

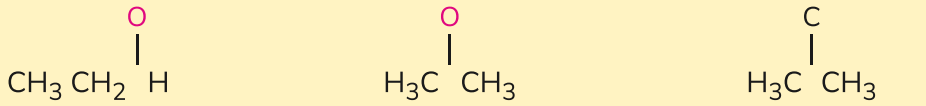
- boiling points of alcohols and phenols **climb with the number of carbon atoms**, as the **van der Waals forces grow**
- among alcohols they **fall as branching in the carbon chain increases**, since a branched molecule has **less surface area** and so weaker van der Waals forces

- the **–OH group** of an alcohol or a phenol takes part in **intermolecular hydrogen bonding**, drawn below





- alcohols and phenols **boil higher than other classes of comparable molecular mass** — hydrocarbons, ethers, haloalkanes and haloarenes among them
- **ethanol and propane** have comparable molecular masses, yet their boiling points are **far apart**
- the boiling point of **methoxymethane falls between the two**



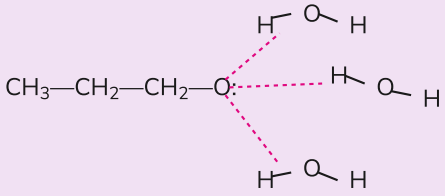
Ethanol · 46 / 351 K *Methoxymethane · 46 / 248 K* *Propane · 44 / 231 K*

molecular mass / boiling point

- alcohols boil high mainly because of the **intermolecular hydrogen bonding** within them, which **ethers and hydrocarbons lack**

Solubility

- alcohols and phenols dissolve in water because they can **form hydrogen bonds with water molecules**
- solubility **falls as the alkyl or aryl (hydrophobic) group grows**
- several of the **lower molecular mass alcohols mix with water in all proportions**



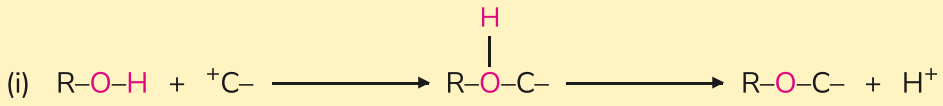
Worked example 7.c

Put each set in order of rising boiling point.

- (a) Pentan-1-ol, butan-1-ol, butan-2-ol, ethanol, propan-1-ol, methanol
- (b) Pentan-1-ol, *n*-butane, pentanal, ethoxyethane
- **(a)** Methanol, ethanol, propan-1-ol, butan-2-ol, butan-1-ol, pentan-1-ol
- **(b)** *n*-Butane, ethoxyethane, pentanal and pentan-1-ol

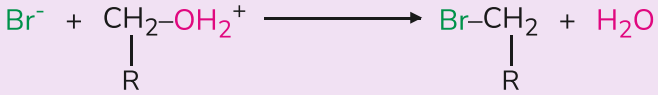
A.6 Chemical reactions

- alcohols are **versatile**: they react **both as nucleophiles and as electrophiles**
- reacting as a **nucleophile**, an alcohol breaks its **O–H bond**



- (ii) reacting as an **electrophile**, an alcohol breaks its **C–O bond**; it is **protonated alcohols** that behave this way



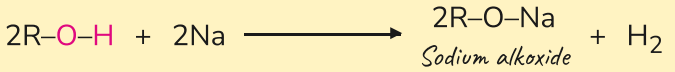


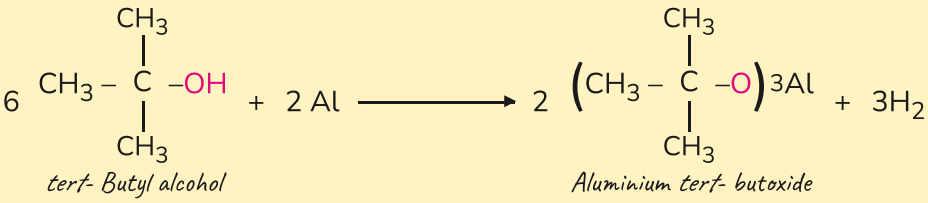
- taking the **cleavage of the O–H and the C–O bonds** as the dividing line, the reactions of alcohols and phenols fall into **two groups** — set out on sheet 13

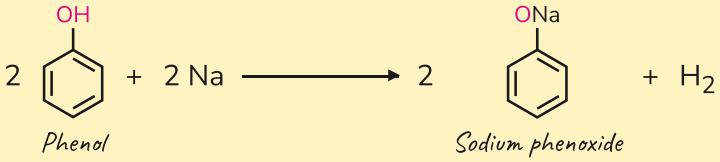
(a) Reactions that cleave the O–H bond

1 · Acidity of alcohols and phenols

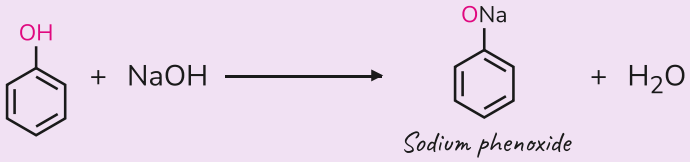
- **(i) With metals** — set an alcohol or a phenol against an **active metal** — sodium, potassium, aluminium — and the matching **alkoxide or phenoxide** results, with **hydrogen** released



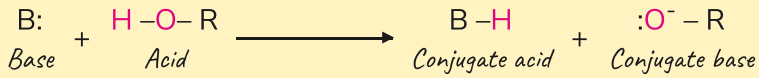




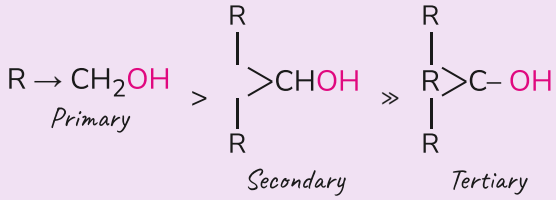
- phenols go further: they react with **aqueous sodium hydroxide** to give **sodium phenoxides**



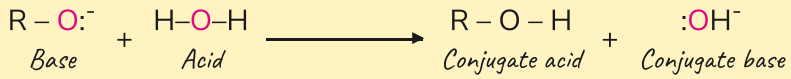
- reactions of that kind mark alcohols and phenols out as **acidic**
- they are in fact **Brönsted acids**: they can **donate a proton to a stronger base (B:)**



- **(ii) Acidity of alcohols** — an alcohol's acidic character rests on the **polar nature of its O–H bond**
- an **electron-releasing group** (–CH_3, –C_2H_5) **raises the electron density on oxygen**, which tends to **lower the polarity of the O-H bond** and so **weakens the acid**
- acid strength therefore falls along this order

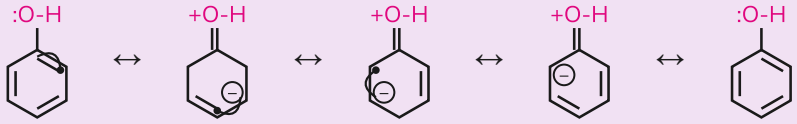


- alcohols are nevertheless **weaker acids than water**, as the reaction of water with an alkoxide shows

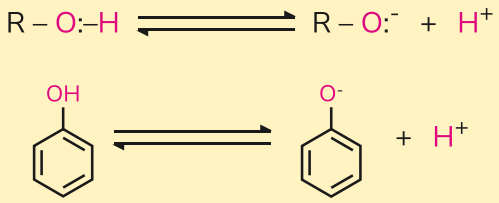


- so **water is the better proton donor**, that is, the stronger acid
- an **alkoxide ion takes up a proton more readily than a hydroxide ion**, so **alkoxides are the stronger bases**: put sodium ethoxide above sodium hydroxide
- alcohols also act as **Bronsted bases**, the **unshared electron pairs on oxygen** making them proton acceptors

- **(iii) Acidity of phenols** — phenol's reactions with metals (sodium, aluminium) and with sodium hydroxide mark it out as **acidic**
- a phenol's hydroxyl group sits straight on a benzene ring's ***sp*^{2} hybridised carbon**, and the ring **withdraws electrons**
- the charge distribution that follows, set out in the resonance structures below, leaves the **oxygen of the –OH group positive**

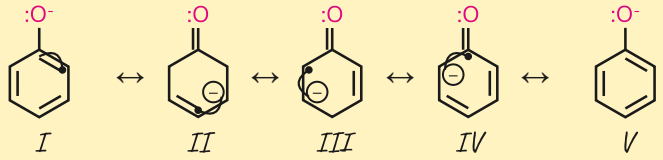


- phenol's reaction with aqueous sodium hydroxide marks phenols out as **stronger acids than alcohols and than water**
- so why is a hydroxyl group on an **aromatic ring** more acidic than one on an **alkyl group**? Compare how the two ionise



- the ***sp*^{2} hybridised carbon** the –OH is attached to in phenol is the **more electronegative**, so it **draws electron density off the oxygen**
- that **raises the polarity of the O–H bond** and leaves **phenols more ionised than alcohols**
- compare the two ions: an **alkoxide** holds its negative charge **on the oxygen alone**, a **phenoxide** spreads it out
- that delocalisation, across structures I-V, is what makes the difference — continued on sheet 15

- delocalisation makes the **phenoxide ion the more stable**, and so **favours the ionisation of phenol**
- charge is delocalised in phenol as well, but its resonance structures carry **separated charges**, which leaves the **phenol molecule less stable than the phenoxide ion**



- in a **substituted phenol**, an **electron withdrawing group** such as a nitro group **raises the acid strength**
- the effect is **more pronounced with such a group at the *ortho* and *para* positions**, where the negative charge of the phenoxide ion is **delocalised effectively**
- **electron releasing groups**, alkyl groups among them, generally **work against the phenoxide ion**, so the acid weakens: put the **cresols below phenol**

- a higher pK_a means a weaker acid

Table 7.C pK_a Values of some Phenols and Ethanol

Compound	Formula	pK_a	Compound	Formula	pK_a
o-Nitrophenol	o–O_2N–C_6H_4–OH	7.2	*o*-Cresol	o–CH_3–C_6H_4–OH	10.2
m-Nitrophenol	m–O_2N–C_6H_4–OH	8.3	*m*-Cresol	m–$CH_3C_6H_4$–OH	10.1
p-Nitrophenol	p–O_2N–C_6H_4–OH	7.1	*p*-Cresol	p–CH_3–C_6H_4–OH	10.2
Phenol	C_6H_5–OH	10.0	Ethanol	C_2H_5OH	15.9

- reading those figures, **phenol is a million times more acidic than ethanol**

Worked example 7.d

Put these compounds in order of rising acid strength: propan-1-ol, 2,4,6-trinitrophenol, 3-nitrophenol, 3,5-dinitrophenol, phenol, 4-methylphenol.

- Propan-1-ol, 4-methylphenol, phenol, 3-nitrophenol, 3,5-dinitrophenol, 2,4, 6-trinitrophenol

2 · Esterification

- set an alcohol or a phenol against a **carboxylic acid, an acid chloride or an acid anhydride** and an **ester** forms

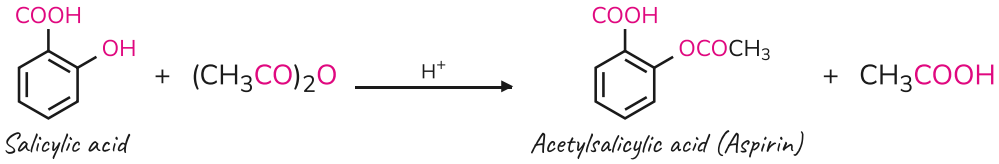
$$Ar/ROH + R'-COOH \underset{}{\overset{H^+}{\rightleftharpoons}} Ar/ROCOR' + H_2O$$

$$Ar/R-OH + (R'CO)_2O \overset{H^+}{\rightleftharpoons} Ar/ROCOR + R'COOH$$

$$R/ArOH + R'COCl \xrightarrow{\text{Pyridine}} R/ArOCOR' + HCl$$

- with a **carboxylic acid or an acid anhydride** the reaction is run in a **small amount of concentrated sulphuric acid**
- it is **reversible**, so the **water is removed as soon as it forms**
- with an **acid chloride** the reaction is run in a base, **pyridine**, which **neutralises the HCl** given off and so **shifts the equilibrium to the right**
- putting an **acetyl (CH_3CO) group** into an alcohol or a phenol is called **acetylation**; acetylating **salicylic acid produces aspirin**

- aspirin relieves pain, damps inflammation and brings fever down



(b) Reactions that cleave the carbon–oxygen (C–O) bond in alcohols

- the C–O bond is broken **in alcohols only**; a **phenol enters such a reaction with zinc and nothing else**

1 · Reaction with hydrogen halides

- a **hydrogen halide** turns an alcohol into an **alkyl halide** (Unit 6, Class XII)

$$ROH + HX \longrightarrow R-X + H_2O$$

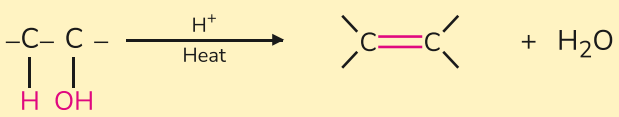
- the **three classes of alcohol differ in how fast they react with HCl**, which tells them apart — the **Lucas test**
- alcohols dissolve in **Lucas reagent** (concentrated HCl with $ZnCl_2$) while their **halides do not**, so the halide clouds the solution
- a **tertiary** alcohol clouds it **at once**, its halide forming readily; a **primary** one leaves the solution **clear at room temperature**

2 · Reaction with phosphorus trihalides

- **phosphorus tribromide** takes an alcohol to the **alkyl bromide** (Unit 6, Class XII)

3 · Dehydration

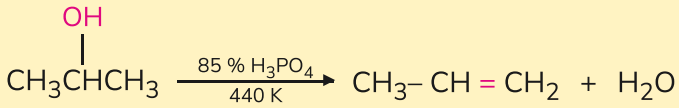
- alcohols lose a molecule of water — they **dehydrate** — to give **alkenes**, on treatment with a **protic acid** such as concentrated H_2SO_4 or H_3PO_4, or with catalysts such as **anhydrous zinc chloride or alumina**

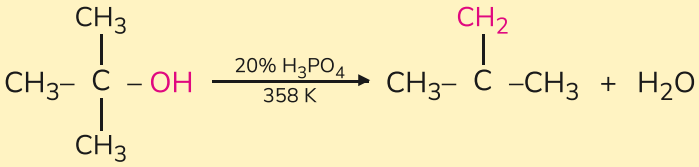


- ethanol dehydrates on **heating with concentrated H_2SO_4 at 443 K**

$$C_2H_5OH \xrightarrow[443\ K]{H_2SO_4} CH_2 = CH_2 + H_2O$$

- **secondary and tertiary alcohols dehydrate under milder conditions**

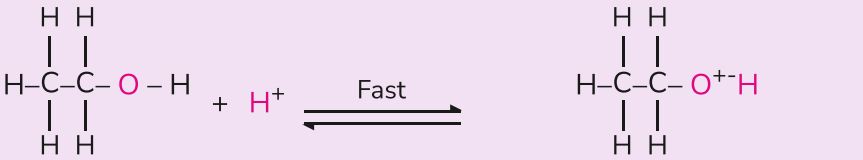




- so the **ease of dehydration** runs **tertiary > secondary > primary**

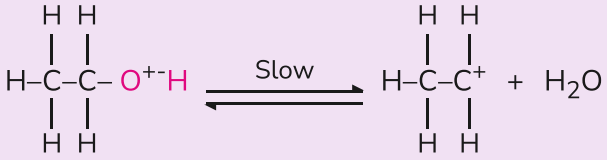
Mechanism · dehydration of ethanol

- **Step 1** — the alcohol is **protonated**

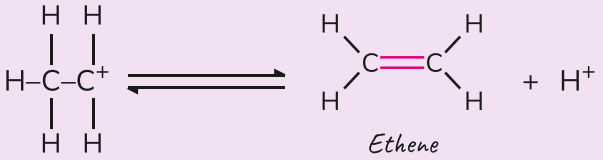


Ethanol *Protonated alcohol (Ethyl oxonium ion)*

- **Step 2** — a **carbocation** forms; this is the **slowest step**, and so the **rate determining** one



- **Step 3** — **a proton is eliminated** and ethene results



Ethene

- tertiary carbocations are more stable, and so form more easily than secondary and primary ones; tertiary alcohols are therefore the easiest to dehydrate

- the acid taken up in step 1 is **released again in step 3**; to drive the equilibrium to the right, the **ethene is removed as it forms**

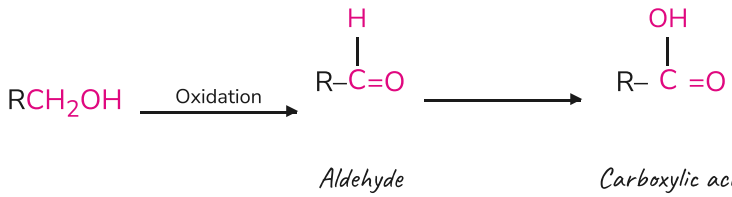
4 · Oxidation

- oxidising an alcohol builds a **carbon-oxygen double bond**, breaking an **O-H and a C-H bond** in the process



Bond breaking

- because bonds are both broken and formed, such reactions are also called **dehydrogenation** reactions: dihydrogen is lost from the alcohol molecule
- according to the oxidising agent, a **primary alcohol is oxidised to an aldehyde**, and that aldehyde in turn to a **carboxylic acid**



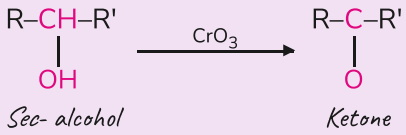
- a **strong oxidising agent**, **potassium permanganate under acid** for one, takes an alcohol straight to the **carboxylic acid**
- **CrO_3 in an anhydrous medium** is the oxidising agent used when the **aldehyde** is to be isolated

$$RCH_2OH \xrightarrow{CrO_3} RCHO$$

- a better reagent, and a good yield with it, is **pyridinium chlorochromate (PCC)**, in which **chromium trioxide is complexed with pyridine and HCl**

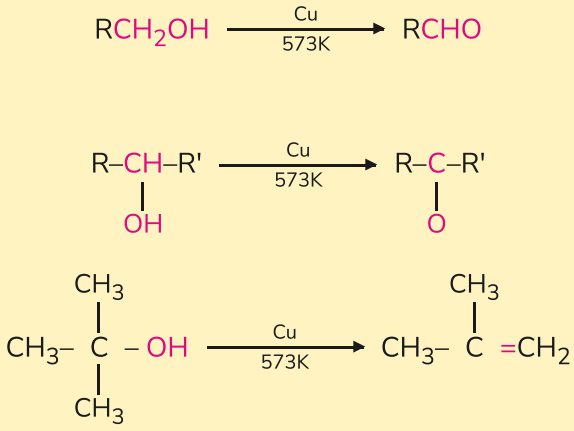
$$CH_3 - CH = CH - CH_2OH \xrightarrow{PCC} CH_3 - CH = CH - CHO$$

- **secondary alcohols are oxidised to ketones by chromic anhydride (CrO_3)**



- **tertiary alcohols do not undergo oxidation**
- under strong conditions — a strong oxidising agent such as $KMnO_4$, and a raised temperature — **various C-C bonds are cleaved**, and a **mixture of carboxylic acids of fewer carbon atoms** results

- pass the vapours of a **primary or a secondary alcohol over heated copper at 573 K** and **dehydrogenation** follows, giving an **aldehyde or a ketone**
- **tertiary alcohols dehydrate** instead



- **biological oxidation** of methanol and of ethanol in the body yields first the matching **aldehyde** and then the **acid**
- alcoholics sometimes drink ethanol mixed by mistake with methanol, that is **denatured alcohol**
- the body oxidises **methanol first to methanal, then to methanoic acid**, which can leave a person **blind or dead**
- the treatment for methanol poisoning is an **intravenous infusion of diluted ethanol**: it **swamps the enzyme** that would oxidise the aldehyde (HCHO) to the acid, buying the kidneys time to excrete the methanol

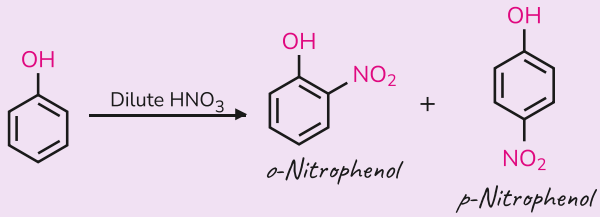
(c) Reactions of phenols

- the reactions that follow on sheet 19 are shown by **phenols only**

1 · Electrophilic aromatic substitution

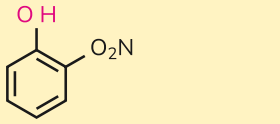
- the reactions phenols undergo on the aromatic ring are **electrophilic substitutions** (Unit 9, Class XI)
- the –OH group makes its benzene ring **readier to take an electrophile**
- it also **directs the incoming group to the *ortho* and *para* positions**, which the **resonance effect of –OH leaves rich in electrons**
- the resonance structures are those drawn under the acidity of phenols

- **(i) Nitration — dilute nitric acid at a low temperature (298 K)** takes phenol to a **mixture of *ortho* and *para* nitrophenols**

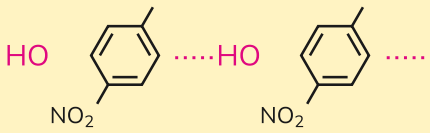


o-Nitrophenol *p*-Nitrophenol

- the ***ortho* and *para* isomers separate by steam distillation**
- ***o*-Nitrophenol is steam volatile**, its hydrogen bonding being **intramolecular**
- ***p*-nitrophenol is less volatile**: its hydrogen bonding is **intermolecular**, and that **associates the molecules**



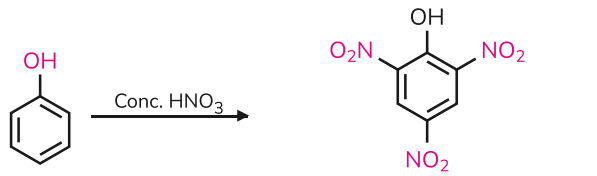
o-Nitrophenol (Intramolecular H-bonding)



p-Nitrophenol (Intermolecular H-bonding)

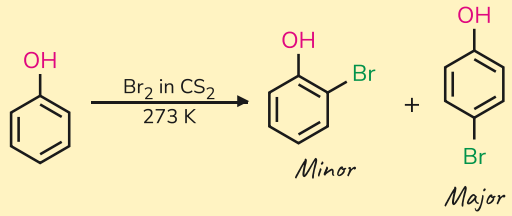
- **concentrated nitric acid** takes phenol to **2,4,6-trinitrophenol**, commonly known as **picric acid**
- the **yield of this product is poor**
- picric acid is now made in two stages: **concentrated sulphuric acid** first, which gives **phenol-2,4-disulphonic acid**, and **concentrated nitric acid** after, which carries it to 2,4,6-trinitrophenol

- 2, 4, 6 - trinitrophenol is a strong acid: its three electron withdrawing $–NO_2$ groups make the release of a hydrogen ion easy

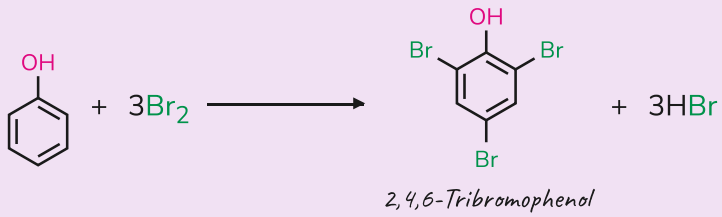


2,4,6-Trinitrophenol (Picric acid)

- **(ii) Halogenation** — treat phenol with bromine and **different conditions give different products**
- **(a)** keep the polarity of the solvent low — $CHCl_3$, CS_2 — and the temperature down, and **monobromophenols** are what form



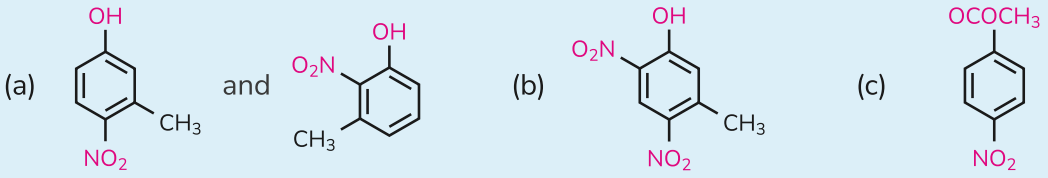
- to halogenate benzene one usually needs **$FeBr_3$ or another Lewis acid** (Unit 6, Class XII) to **polarise the halogen molecule**
- with phenol the **bromine polarises even without a Lewis acid**, because the **–OH group on the ring is so strongly activating**
- **(b)** treat phenol with **bromine water** and **2,4,6-tribromophenol** comes down as a **white precipitate**



Worked example 7.e

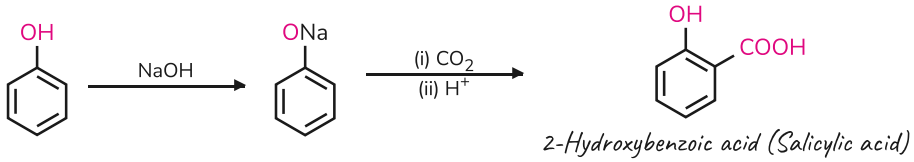
Draw the structure of the major product expected from each reaction: (a) mononitration of 3-methylphenol; (b) dinitration of 3-methylphenol; (c) mononitration of phenyl methanoate.

- where the incoming group lands is settled by the **–OH and the –CH_3 group acting together**



2 · Kolbe's reaction

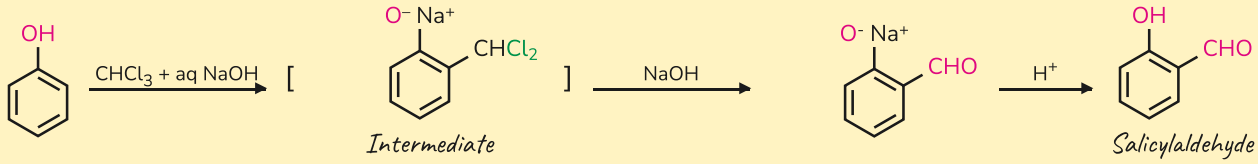
- sodium hydroxide turns phenol into the **phenoxide ion**, which towards electrophilic aromatic substitution is **more reactive still than phenol**
- even **carbon dioxide**, a **weak electrophile**, will substitute into it
- the main product is ***ortho* hydroxybenzoic acid**



2-Hydroxybenzoic acid (Salicylic acid)

3 · Reimer-Tiemann reaction

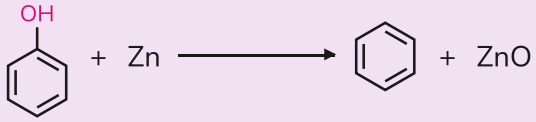
- put **chloroform and sodium hydroxide** to phenol and a **–CHO group takes the *ortho* position** on the ring; that is the **Reimer - Tiemann reaction**
- **alkali then hydrolyses** the intermediate **substituted benzal chloride**, and **salicylaldehyde** results



Intermediate Salicylaldehyde

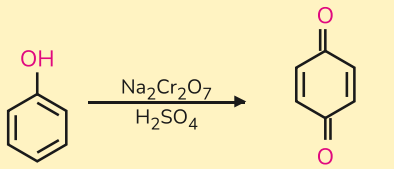
4 · Reaction of phenol with zinc dust

- **heating phenol with zinc dust converts it to benzene**



5 · Oxidation

- oxidising phenol with **chromic acid** yields a **conjugated diketone** known as **benzoquinone**
- left in air, phenols **oxidise slowly to dark mixtures that hold quinones**



benzoquinone

Intext questions

Q6 Draw the products expected from each alcohol below with (a) HCl –$ZnCl_2$, (b) HBr and (c) $SOCl_2$: (i) butan-1-ol, (ii) 2-methylbutan-2-ol.

Q7 Predict the major product of the acid catalysed dehydration of (i) 1-methylcyclohexanol and (ii) butan-1-ol.

Q8 Both the *ortho* and the *para* nitrophenol beat phenol for acidity. Draw the resonance structures of their phenoxide ions.

Q9 Write the equations of (i) the Reimer - Tiemann reaction and (ii) Kolbe's reaction.

A.7 Some commercially important alcohols

- **methanol and ethanol** are the two alcohols that matter commercially

1 · Methanol

- methanol, CH_3OH, went by the name **'wood spirit'**, and was once got by **distilling wood destructively**
- most of it now comes from **carbon monoxide hydrogenated catalytically**, under **high pressure and temperature over ZnO – Cr_2O_3**

$$CO + 2H_2 \xrightarrow[200\text{–}300\ \text{atm} \cdot 573\text{–}673\ \text{K}]{ZnO\text{–}Cr_2O_3} CH_3OH$$

- it is a **colourless liquid boiling at 337 K**, and it is **highly poisonous**
- swallowing even a **small quantity causes blindness**, and a **large quantity death**
- it serves as a **solvent in paint and varnish**, and above all for **making formaldehyde**

2 · Ethanol

- ethanol, C_2H_5OH, comes commercially from **fermentation**; the oldest way of all begins **with sugar**
- the enzyme **invertase** turns the sugar of **molasses, of sugarcane, of a fruit such as the grape** into **glucose and fructose**, each $C_6H_{12}O_6$
- glucose and fructose then **ferment** under another enzyme, **zymase**, which is found in **yeast**

- ethanol swallowed goes to work on the **central nervous system**: a moderate amount clouds judgment and loosens inhibition; more brings nausea and then unconsciousness; more still disturbs breathing itself, and can kill

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{Invertase}} \underset{\textit{Glucose}}{C_6H_{12}O_6} + \underset{\textit{Fructose}}{C_6H_{12}O_6}$$

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} 2C_2H_5OH + 2CO_2$$

- in **wine making** the grape supplies both sugar and yeast: ripening **builds the sugar up**, while **yeast settles on the skin outside**
- crushing the grapes brings sugar and enzyme into contact, and **fermentation begins**
- fermentation runs under **anaerobic conditions**, that is in the absence of air, and **carbon dioxide is released**
- zymase is **inhibited once the alcohol formed passes 14 percent**
- should air reach the mixture, its **oxygen oxidises the ethanol to ethanoic acid**, which ruins the taste of the drink

- ethanol is a **colourless liquid boiling at 351 K**
- it serves as a **solvent in the paint industry** and in preparing a great many carbon compounds
- commercial alcohol is made **unfit to drink** by mixing in a little **copper sulphate**, which colours it, and **pyridine**, which smells foul; this is the **denaturation** of alcohol
- large quantities of ethanol are now obtained by the **hydration of ethene** (section A.4)

B.1 Ethers · preparation

1 · By dehydration of alcohols

- alcohols dehydrate in the presence of **protic acids** (H_2SO_4, H_3PO_4)
- whether the product is an **alkene or an ether** turns on the **reaction conditions**
- ethanol with sulphuric acid gives **ethene at 443 K**, but **ethoxyethane is the main product at 413 K**

- diethyl ether was once widely used as an **inhalation anaesthetic**, but its slow effect and unpleasant recovery period have seen other compounds replace it

$$CH_3CH_2OH \xrightarrow[443\ K]{H_2SO_4} CH_2{=}CH_2 \qquad CH_3CH_2OH \xrightarrow[413\ K]{H_2SO_4} C_2H_5OC_2H_5$$

- ether formation is a **nucleophilic bimolecular reaction (S_N2)**: an **alcohol molecule attacks a protonated alcohol**

(i) $CH_3{-}CH_2{-}O{-}H + H^+ \longrightarrow CH_3{-}CH_2{-}\overset{\overset{H}{|}}{O}{-}H$

(ii) $CH_3CH_2{-}\overset{\overset{H}{|}}{O}{:} + CH_3{-}CH_2{-}\overset{\overset{H}{|}}{O^+}{-}H \longrightarrow CH_3CH_2{-}\overset{\overset{H}{|}}{O^+}{-}CH_2CH_3 + H_2O$

(iii) $CH_3CH_2{-}\overset{\overset{H}{|}}{O^+}{-}CH_2CH_3 \longrightarrow CH_3CH_2{-}O{-}CH_2CH_3 + H^+$

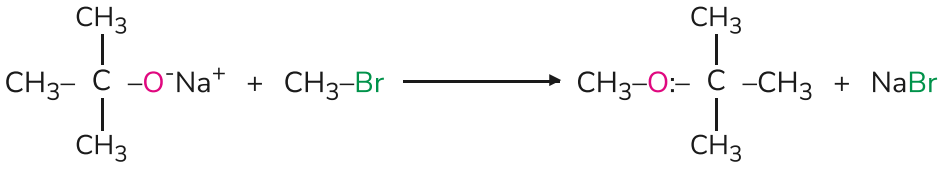
- acidic dehydration of an alcohol to an **alkene** always carries with it a **substitution** that gives an **ether**
- the method suits **primary alkyl groups only**: the alkyl group must be **unhindered** and the **temperature kept low**, or the reaction turns to the alkene
- with a **secondary or tertiary alcohol** the reaction follows the **S_N1 pathway**
- dehydrating a secondary or tertiary alcohol to the matching ether **does not succeed**: **elimination competes with substitution**, and alkenes form readily
- can you say why bimolecular dehydration will not serve for preparing ethyl methyl ether?

2 · Williamson synthesis

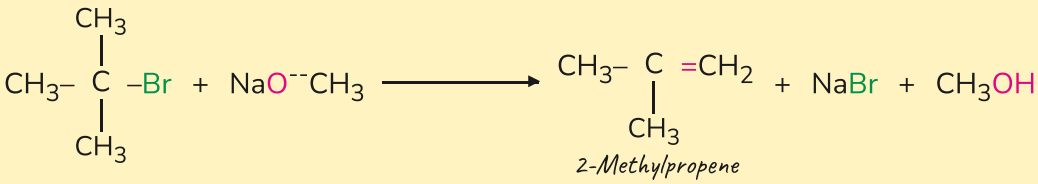
- an important laboratory route to **ethers symmetrical and unsymmetrical alike**: set an **alkyl halide against a sodium alkoxide**
- ethers whose alkyl groups are **substituted**, secondary or tertiary, come the same way; what happens is an **S_N2 attack by an alkoxide ion on a primary alkyl halide**

$$R{-}X + R'{-}O{:}^-Na^+ \longrightarrow R{-}O{:}{-}R' + Na\,X$$

- Alexander William Williamson (1824-1904), born in London to Scottish parents, took the Chair of Chemistry at University College, London, in 1849



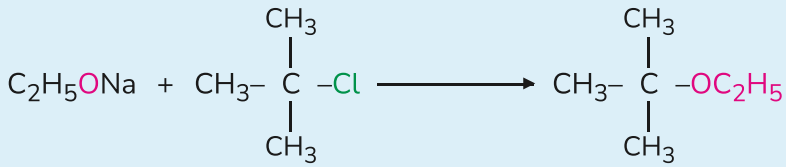
- the results are **better where the alkyl halide is primary**
- with **secondary and tertiary alkyl halides, elimination competes with substitution**
- use a **tertiary alkyl halide** and an **alkene is the only product**: no ether forms at all
- CH_3ONa with $(CH_3)_3C–Br$ gives **2-methylpropene and nothing else**



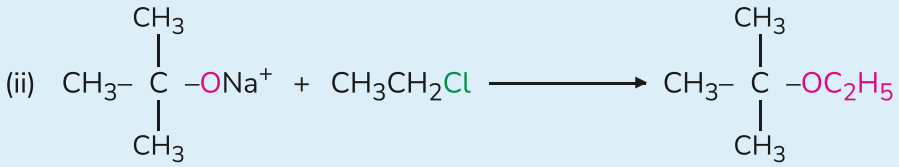
- the reason: **alkoxides are strong bases as well as nucleophiles**, and against alkyl halides that leads to **elimination**

Worked example 7.f

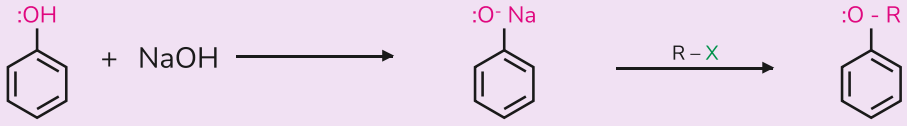
This is not an appropriate reaction for preparing t-butyl ethyl ether.



- (i) what would the major product of this reaction be?
- (ii) write a suitable reaction for preparing t-butylethyl ether
- **(i)** the major product is **2-methylprop-1-ene**: sodium ethoxide is a **strong base as well as a strong nucleophile**, so **elimination predominates over substitution**



- **phenols too are converted to ethers** by this method, phenol serving as the **phenoxide moiety**



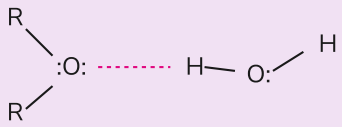
B.2 Physical properties

- the **C-O bonds in an ether are polar**, so the molecule carries a **net dipole moment**
- that polarity is **weak**, and does not tell much on the boiling points: they are **comparable to those of alkanes of similar molecular mass**, but **far below those of alcohols**

Formula	$CH_3(CH_2)_3CH_3$ *n*-Pentane	$C_2H_5–O–C_2H_5$ Ethoxyethane	$CH_3(CH_2)_3–OH$ Butan-1-ol
b.p./K	309.1	307.6	390

- the wide gap between alcohols and ethers comes from the **hydrogen bonding present in alcohols**

- an ether **mixes with water about as well as an alcohol of the same molecular mass** does
- **ethoxyethane and butan-1-ol** are miscible to nearly the same extent — **7.5 and 9 g per 100 mL of water**
- **pentane is essentially immiscible** with water
- the reason: like an alcohol, the **oxygen of an ether can form hydrogen bonds with water molecules**



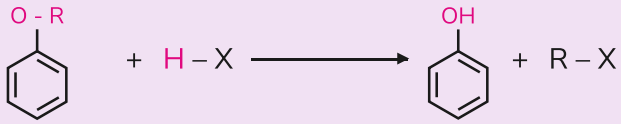
B.3 Chemical reactions

1 · Cleavage of the C–O bond in ethers

- of all the functional groups, the **ether is the least reactive**
- their C-O bond is cleaved only under **drastic conditions, with an excess of hydrogen halide**
- a **dialkyl ether** so treated gives **two alkyl halide molecules**

$$R–O–R + HX \longrightarrow RX + R–OH \qquad R–OH + HX \longrightarrow R–X + H_2O$$

- an **alkyl aryl ether is cleaved at the alkyl-oxygen bond**, the **aryl-oxygen bond being the more stable**
- the products are a **phenol and an alkyl halide**



- an ether carrying **two different alkyl groups** is cleaved the same way

$$R–O–R' + HX \longrightarrow R–X + R'–OH$$

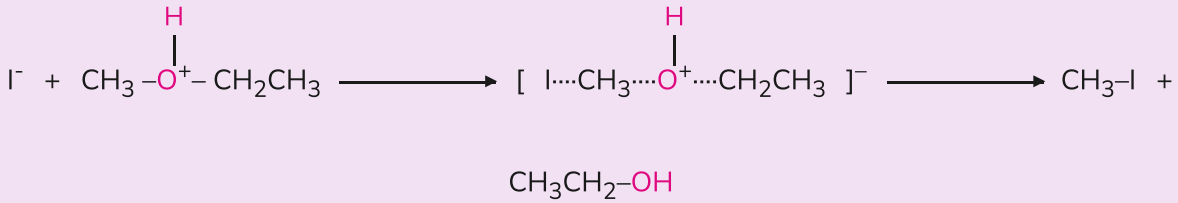
- the hydrogen halides rank **HI > HBr > HCl** in reactivity
- the cleavage is carried out with **concentrated HI or HBr at high temperature**

Mechanism

- an ether met by **concentrated HI** is **protonated** first
- **Step 1**



- only **HBr and HI** serve here; they alone are **acidic enough**
- **Step 2 — iodide makes a good nucleophile**: it goes for the **carbon that carries fewest substituents** in the oxonium ion of step 1, and **drives out an alcohol molecule, S_N2 fashion**
- so with a **mixed ether**, which alcohol and which alkyl iodide result **turns on the nature of the alkyl groups**
- where the groups are **primary or secondary**, it is the **lower alkyl group that forms the alkyl iodide** (an S_N2 reaction)

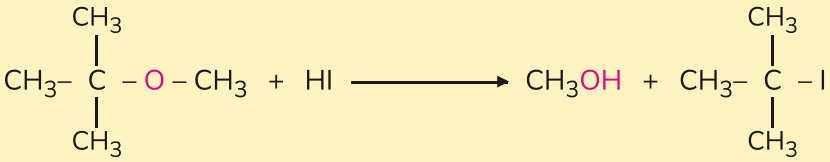


- with **HI in excess and the temperature high**, the ethanol **meets another HI molecule** and becomes **ethyl iodide**
- **Step 3**

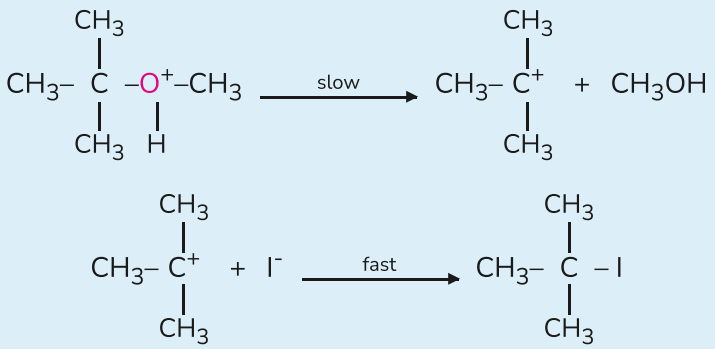




- where **one alkyl group is tertiary**, however, **the halide formed is the tertiary halide**



- the reason: when $HO–CH_3$ leaves in step 2 it leaves behind the **more stable carbocation**, $[(CH_3)_3C^+]$, and the **S_N1 path is taken**

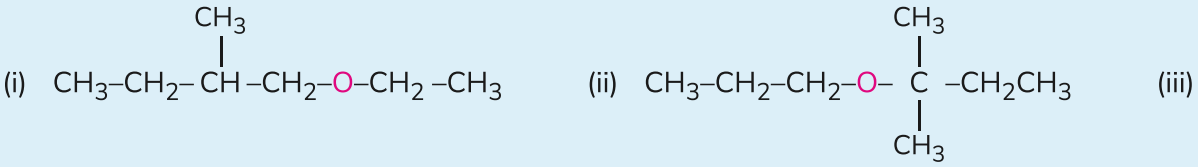


- with **anisole**, protonation of the ether gives the **methylphenyl oxonium ion**, $C_6H_5–O^+(H)–CH_3$
- the **O–CH_3 bond is weaker than the O–C_6H_5 bond**: the carbon of the phenyl group is ***sp*^{2} hybridised**, and the bond has **partial double bond character**

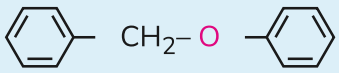
- so the **I^- ion breaks the O–CH_3 bond** and CH_3I results
- **phenols do not react further to give halides**: the sp^2 hybridised carbon of a phenol **cannot undergo the nucleophilic substitution** that conversion to the halide would need

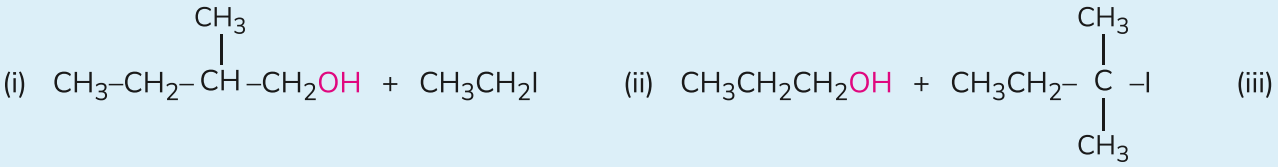
Worked example 7.g

Give the major products formed by heating each ether below with HI.

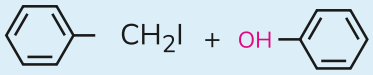


(i) (ii) (iii)



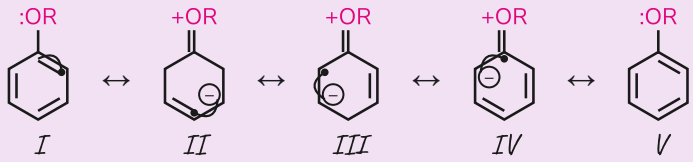


(i) (ii) (iii)

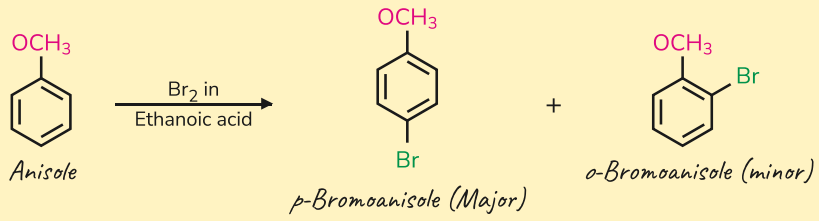


2 · Electrophilic substitution

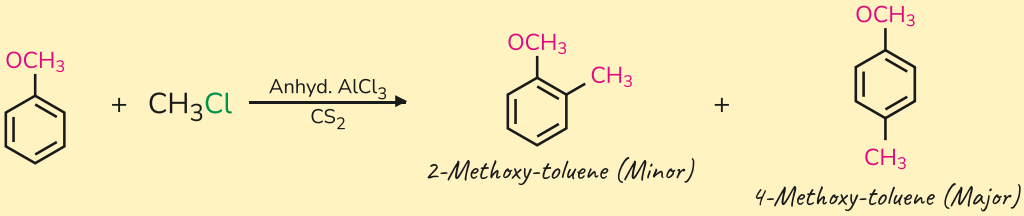
- the **alkoxy group (-OR)** sends an incoming group ***ortho* and *para***, and leaves the **aromatic ring readier to take an electrophile**, exactly as –OH does in phenol

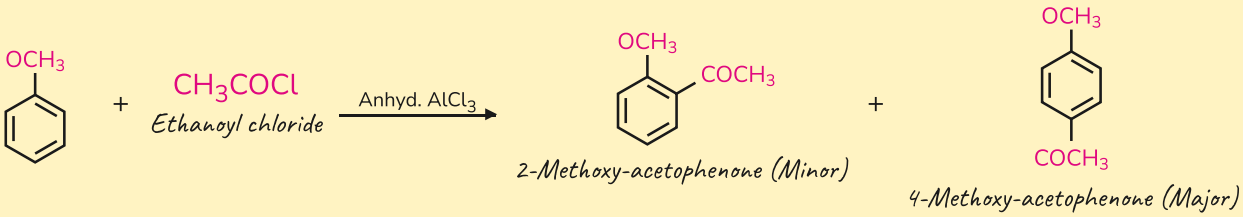


- **(i) Halogenation** — phenylalkyl ethers undergo the **usual halogenation of the benzene ring**
- **anisole** is brominated by **bromine in ethanoic acid even without an iron (III) bromide catalyst**, because the **methoxy group activates the ring**
- the ***para* isomer comes out in 90% yield**



- **(ii) Friedel-Crafts reaction** — anisole enters this reaction too: **alkyl and acyl groups come in at the *ortho* and *para* positions**
- the reagents are an **alkyl halide** and an **acyl halide**, over **anhydrous aluminium chloride**, a Lewis acid, as catalyst





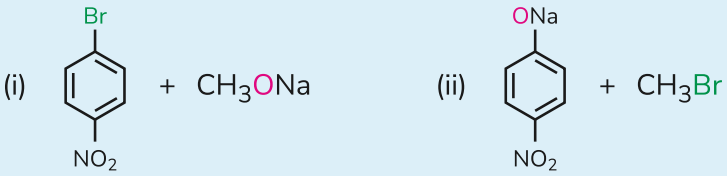
- **(iii) Nitration** — set anisole against **concentrated sulphuric and nitric acids together** and both the ***ortho* and the *para* nitroanisole** result



Intext questions

Q10 Beginning from ethanol and 3-methylpentan-2-ol, write the Williamson synthesis reactions that reach 2-ethoxy-3-methylpentane.

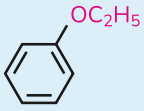
Q11 Which pair below would you choose to prepare 1-methoxy-4-nitrobenzene, and why?

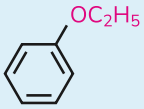


Intext question

Q12 Predict the products of these reactions.

(i) $CH_3 - CH_2 - CH_2 - O - CH_3 + HBr \longrightarrow$

(ii)  $+ HBr \longrightarrow$

(iii)  $\xrightarrow[\text{Conc. } HNO_3]{\text{Conc. } H_2SO_4}$

(iv) $(CH_3)_3C - OC_2H_5 \xrightarrow{HI}$

Summary

- sort **alcohols and phenols** by (i) **how many hydroxyl groups** they carry and (ii) whether the carbon under the –OH is ***sp*^{3} or *sp*^{2}**; sort **ethers** by **what hangs on the oxygen**
- routes to an alcohol: (1) **hydrating an alkene**, whether (i) over acid or (ii) by **hydroboration-oxidation**; (2) from a **carbonyl compound**, whether by (i) **catalytic reduction** or (ii) a **Grignard reagent**
- routes to a phenol: (1) putting **–OH in place of a haloarene's halogen**, or of an aryl sulphonic acid's **sulphonic acid group**; (2) **hydrolysing a diazonium salt**; (3) in industry, **starting from cumene**

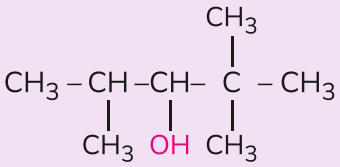
- alcohols **boil higher than other classes of comparable molecular mass** — hydrocarbons, ethers and haloalkanes
- all three families **hydrogen bond to water**, and that is what **dissolves them**
- alcohols and phenols are **acidic**; in a phenol an **electron withdrawing group raises the acid strength** and an **electron releasing group lowers it**

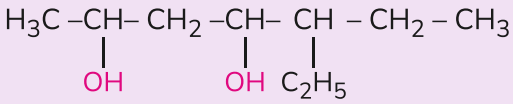
- a hydrogen halide **substitutes nucleophilically** into an alcohol to give an alkyl halide; **dehydration yields an alkene**
- on oxidation, a **primary alcohol yields an aldehyde with a mild oxidising agent** and a **carboxylic acid with a strong one**; a **secondary alcohol yields a ketone**; a **tertiary alcohol resists oxidation**
- a phenol's –OH group **readies the aromatic ring for an electrophile** and, by the resonance effect, **sends the incoming group *ortho* and *para***
- the **Reimer-Tiemann reaction** of phenol yields **salicylaldehyde**
- sodium hydroxide gives the **phenoxide ion**, more reactive still than phenol itself; hence **Kolbe's reaction** in alkaline medium

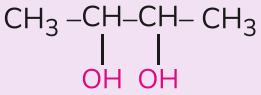
- ethers come (i) by **dehydrating alcohols** and (ii) by the **Williamson synthesis**
- they **boil about where alkanes do**, and **dissolve about as well as alcohols** of the same molecular mass
- a **hydrogen halide cleaves the C–O bond of an ether**
- in electrophilic substitution the **alkoxy group readies the ring** and sends the incoming group to the ***ortho* and *para* positions**

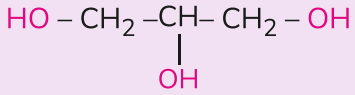
Exercises

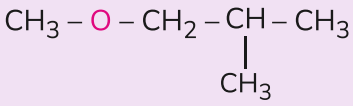
E1 Give each compound below its IUPAC name.

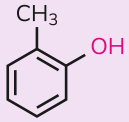
(i) 

(ii) 

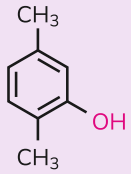
(iii) 

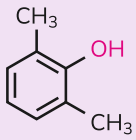
(iv) 

(ix) 

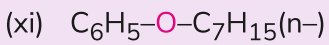
(v) 

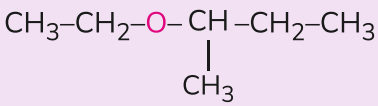
(vi) 

(vii) 

(viii) 

(x) 

(xi) 

(xii) 

E2 Draw the structure of each compound whose IUPAC name is given. (i) 2-Methylbutan-2-ol; (ii) 1-Phenylpropan-2-ol; (iii) 3,5-Dimethylhexane –1, 3, 5-triol; (iv) 2,3 – Diethylphenol; (v) 1 – Ethoxypropane; (vi) 2-Ethoxy-3-methylpentane; (vii) Cyclohexylmethanol; (viii) 3-Cyclohexylpentan-3-ol; (ix) Cyclopent-3-en-1-ol; (x) 4-Chloro-3-ethylbutan-1-ol

E3 (i) Draw every isomeric alcohol of molecular formula $C_5H_{12}O$ and give each its IUPAC name. (ii) Sort those isomers into primary, secondary and tertiary alcohols.

E4 Why does propanol boil higher than the hydrocarbon butane?

E5 Alcohols dissolve in water more readily than hydrocarbons of comparable molecular mass. Account for it.

E6 What does the hydroboration-oxidation reaction mean? Illustrate it.

E7 Which monohydric phenols answer to C_7H_8O? Draw them and name them by IUPAC.

E8 Steam distillation is used to split a mixture of the *ortho* and the *para* nitrophenol. Which isomer comes over, and why?

E9 Give the equations that prepare phenol from cumene.

E10 Write the chemical reaction that prepares phenol from chlorobenzene.

E11 Write the mechanism by which ethene is hydrated to ethanol.

E12 With benzene, concentrated H_2SO_4 and NaOH to hand, write the equations that prepare phenol from them.

E13 Show how you would synthesise: (i) 1-phenylethanol from a suitable alkene; (ii) cyclohexylmethanol from an alkyl halide by an S_N2 reaction; (iii) pentan-1-ol from a suitable alkyl halide.

E14 Give two reactions that display the acidic nature of phenol, and set phenol's acidity beside that of ethanol.

E15 Why is *ortho* nitrophenol a stronger acid than *ortho* methoxyphenol?

E16 How does an –OH group on a carbon of a benzene ring activate that ring towards electrophilic substitution?

E17 Give equations for these reactions. (i) Propan-1-ol oxidised with alkaline $KMnO_4$ solution. (ii) Bromine in CS_2 acting on phenol. (iii) Dilute HNO_3 acting on phenol. (iv) Phenol treated with chloroform in aqueous NaOH.

E18 Take each in turn and explain it, with an example: Kolbe's reaction; the Reimer-Tiemann reaction; the Williamson ether synthesis; an unsymmetrical ether.

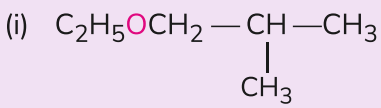
E19 Write the mechanism of the acid dehydration of ethanol to ethene.

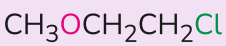
E20 Carry out each of these conversions: propene to propan-2-ol; benzyl chloride to benzyl alcohol; ethyl magnesium chloride to propan-1-ol; methyl magnesium bromide to 2-methylpropan-2-ol.

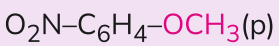
E21 Name the reagent used in each reaction. (i) A primary alcohol oxidised to a carboxylic acid. (ii) A primary alcohol oxidised to an aldehyde. (iii) Phenol brominated to 2,4,6-tribromophenol. (iv) Benzyl alcohol to benzoic acid. (v) Propan-2-ol dehydrated to propene. (vi) Butan-2-one to butan-2-ol.

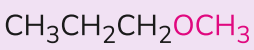
E22 Why does ethanol boil higher than methoxymethane?

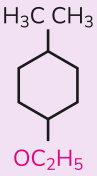
E23 Give each ether below its IUPAC name.

(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

E24 For each ether below, name the reagents and give the equations of a Williamson synthesis: 1-propoxypropane; ethoxybenzene; 2-methoxy-2-methylpropane; 1-methoxyethane.

E25 Illustrate, with examples, where the Williamson synthesis fails for certain types of ether.

E26 How is 1-propoxypropane synthesised from propan-1-ol? Give the mechanism.

E27 Preparing ethers by the acid dehydration of a secondary or tertiary alcohol is not a suitable method. Why?

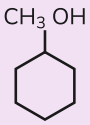
E28 Give the equation when hydrogen iodide acts on each of 1-propoxypropane, methoxybenzene and benzyl ethyl ether.

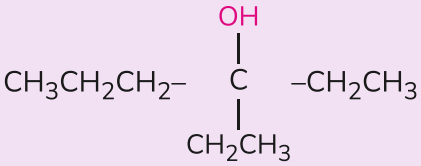
E29 In an aryl alkyl ether, why does the alkoxy group leave the benzene ring readier to take an electrophile, and why does it send the incoming substituent *ortho* and *para*?

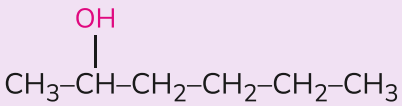
E30 Set out the mechanism when HI acts on methoxymethane.

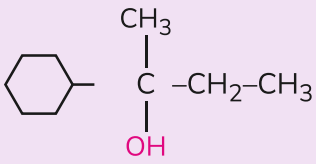
E31 Give equations for each of these: anisole alkylated by Friedel-Crafts; anisole nitrated; anisole brominated in ethanoic acid; anisole acetylated by Friedel-Craft's.

E32 Show how you would synthesise each alcohol below from an appropriate alkene.

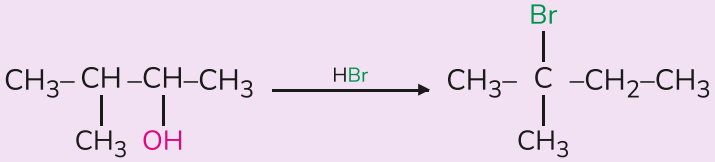
(i) 

(ii) 

(iii) 

(iv) 

E33 Treating 3-methylbutan-2-ol with HBr gives the reaction below. Give a mechanism for it. Hint: a hydride ion shifts from the 3rd carbon, and the secondary carbocation of step II becomes the more stable tertiary one.



Answers to some intext questions

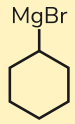
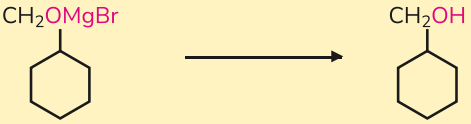
- **Q1** Primary: (i), (ii), (iii). Secondary: (iv), (v). Tertiary: (vi).
- **Q2** Allylic alcohols (ii) and (vi)

- **Q3** (i) 4-Chloro-3-ethyl-2-(1-methylethyl)-butan-1-ol; (ii) 2, 5-Dimethylhexane-1,3-diol; (iii) 3-Bromocyclohexanol; (iv) Hex-1-en-3-ol; (v) 2-Bromo-3-methylbut-2-en-1-ol

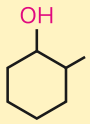
Q4

(i) $CH_3-CH(CH_3)-MgBr + HCHO \longrightarrow CH_3-CH(CH_3)-CH_2-OMgBr \xrightarrow{H_2O}$

$CH_3-CH(CH_3)-CH_2OH + Mg(OH)Br$

(ii) $HCHO +$  

Q5

(i) $CH_3-CH(OH)-CH_3$ (ii)  $CH_2-C(=O)-OCH_3$ (iii) $CH_3-CH_2-CH(CH_3)-CH_2OH$

- **Q7** (i) 1-Methylcyclohexene. (ii) But-1-ene and but-2-ene together; but-2-ene predominates, rearrangement having given the secondary carbocation.

Q10

$CH_3-CH_2-CH(CH_3)-CH(OH)-CH_3 \xrightarrow{Na} CH_3-CH_2-CH(CH_3)-CH(CH_3)-ONa$

$C_2H_5OH \xrightarrow{HBr} C_2H_5Br$

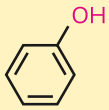
$CH_3-CH_2-CH(CH_3)-CH(CH_3)-ONa + C_2H_5Br \longrightarrow CH_3-CH_2-CH(CH_3)-CH(CH_3)-OC_2H_5$

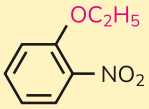
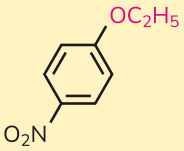
2-Ethoxy-3-methylpentane

- **Q11** Set (ii) is the appropriate pair of reactants.

Q12

(i) $CH_3CH_2CH_2OH + CH_3Br$

(ii)  + C_2H_5Br

(iii)  + 

(iv) $(CH_3)_3\,C - I + C_2H_5OH$